

Nuclear Energy

ENGR49580D Technology, Society and Sustainability

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Uranium Fission Reactors

Uranium Fission Reactors

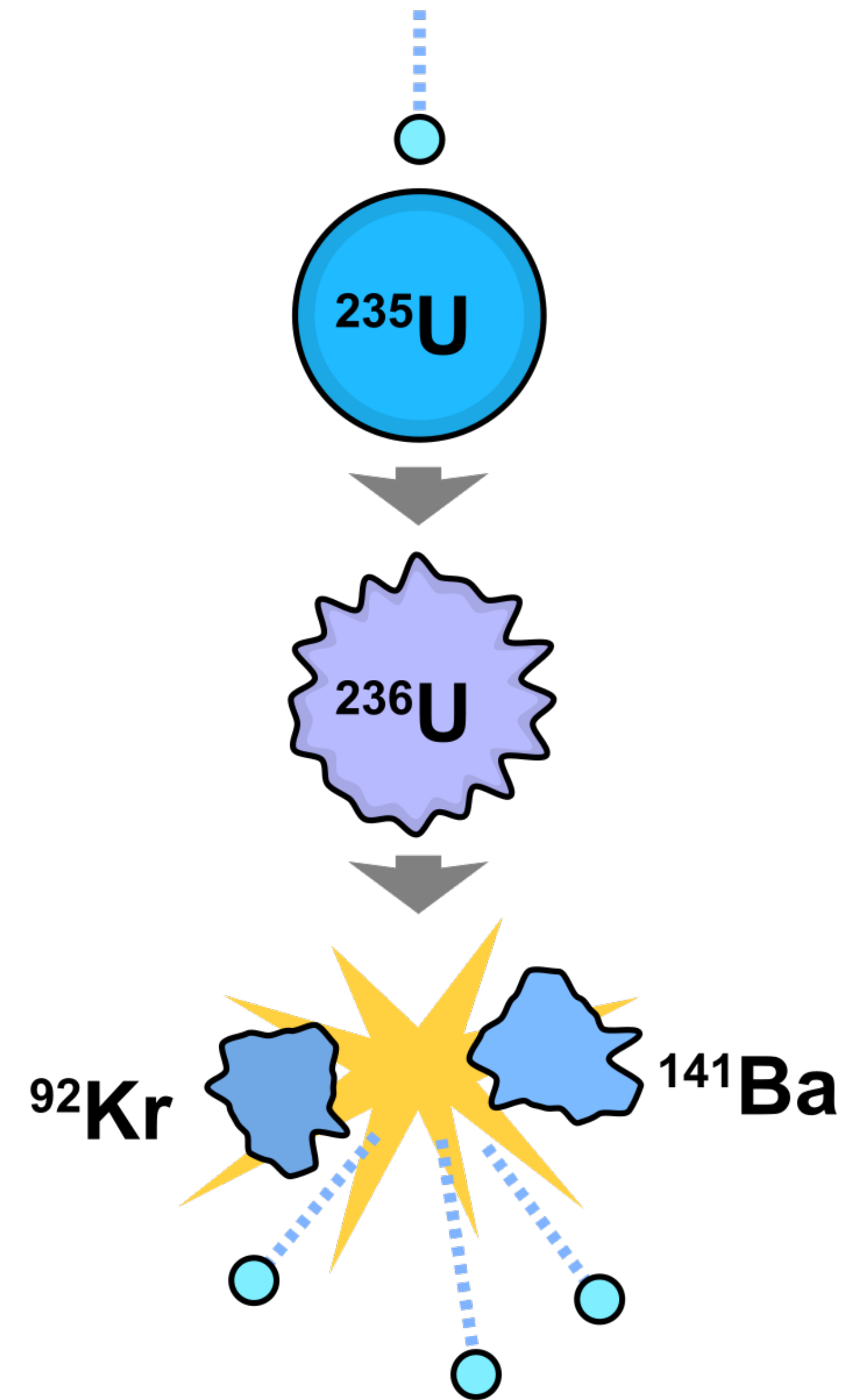
Uses

- Uranium Fission Reactors are the most prevalent reactor design
- They are used for 3 broad purposes:
 - Power Reactors
 - Electrical Power Production / Process Steam Production / Propulsion
 - Research Reactors
 - Neutron Source / Isotope Production (Medical or Industrial)
 - Breeder Reactors
 - Production of fissile fuel from non fissile elements (such as ^{239}Pu from ^{238}U)

Uranium Fission Reactors

The Reaction

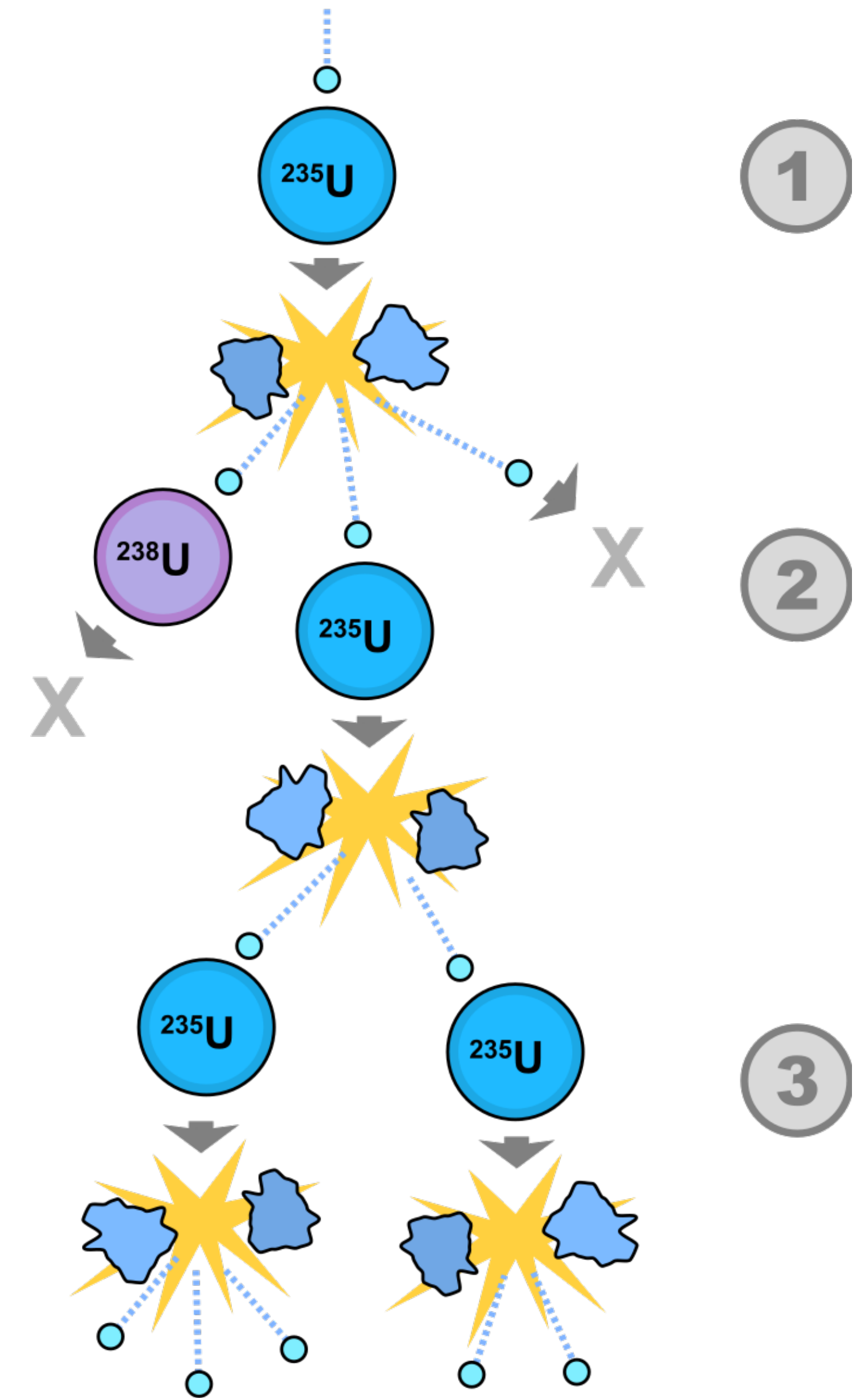
- When a fissionable nucleus captures a free neutron it becomes unstable
- The shape of the nucleus distorts to the point where the range of the nuclear force (the force that bonds protons and neutrons in a nucleus) is exceeded
- This causes the nucleus to split into two smaller nuclei, shedding free neutrons as well as heat



Uranium Fission Reactors

Chain Reactions

- Fissile materials are materials that are able to sustain a fission chain reaction
 - They emit excess neutrons that have sufficient energy to start new fission reactions
- Fissile materials include ^{233}U , ^{235}U and ^{239}Pu
 - ^{235}U is the only naturally occurring isotope
 - ^{233}U is bred from ^{232}Th
 - ^{239}Pu is bred from ^{238}U



Uranium Fission Reactors

Criticality

- Criticality refers to a reactor's ability to sustain a chain reaction
 - In a subcritical assembly, fewer neutrons are produced than are consumed / lost
 - In a critical assembly, the same number of neutrons are produced than are consumed / lost
 - In a supercritical assembly, more neutrons are produced than are consumed / lost

Uranium Fission Reactors

Criticality Control

- A critical assembly is one that achieves a critical reaction
- This may be done using a number of means including
 - Neutron reflectors to reduce neutron loss
 - Moderators to increase the likelihood of neutron absorption
 - Neutron absorbers to reduce the number of available neutrons
 - The mass / geometry of the nuclear fuel (critical mass)

Uranium Fission Reactors

Moderators

- The neutrons released by uranium fission have very high kinetic energy (fast neutrons)
 - Fast neutrons have a much lower probability of being absorbed than thermal (slow) neutrons
- A neutron moderator interacts with fast neutrons to reduce their energy
 - Common moderators include
 - Water (H_2O)
 - Heavy Water (D_2O)
 - Carbon (graphite)

Uranium Fission Reactors

Reactor Poisons (Neutron Absorbers)

- Many elements have high neutron capture cross sections and can remove neutrons from a reactor
 - Reactor poisons that are fission products limit the life of reactor fuel
 - Control rods (made of a reactor poison) are used to control or stop reactions
 - Control rods are used to shape the reaction balancing energy levels throughout the core
 - Control rods may also be used to stop a reactor

Uranium Fission Reactors

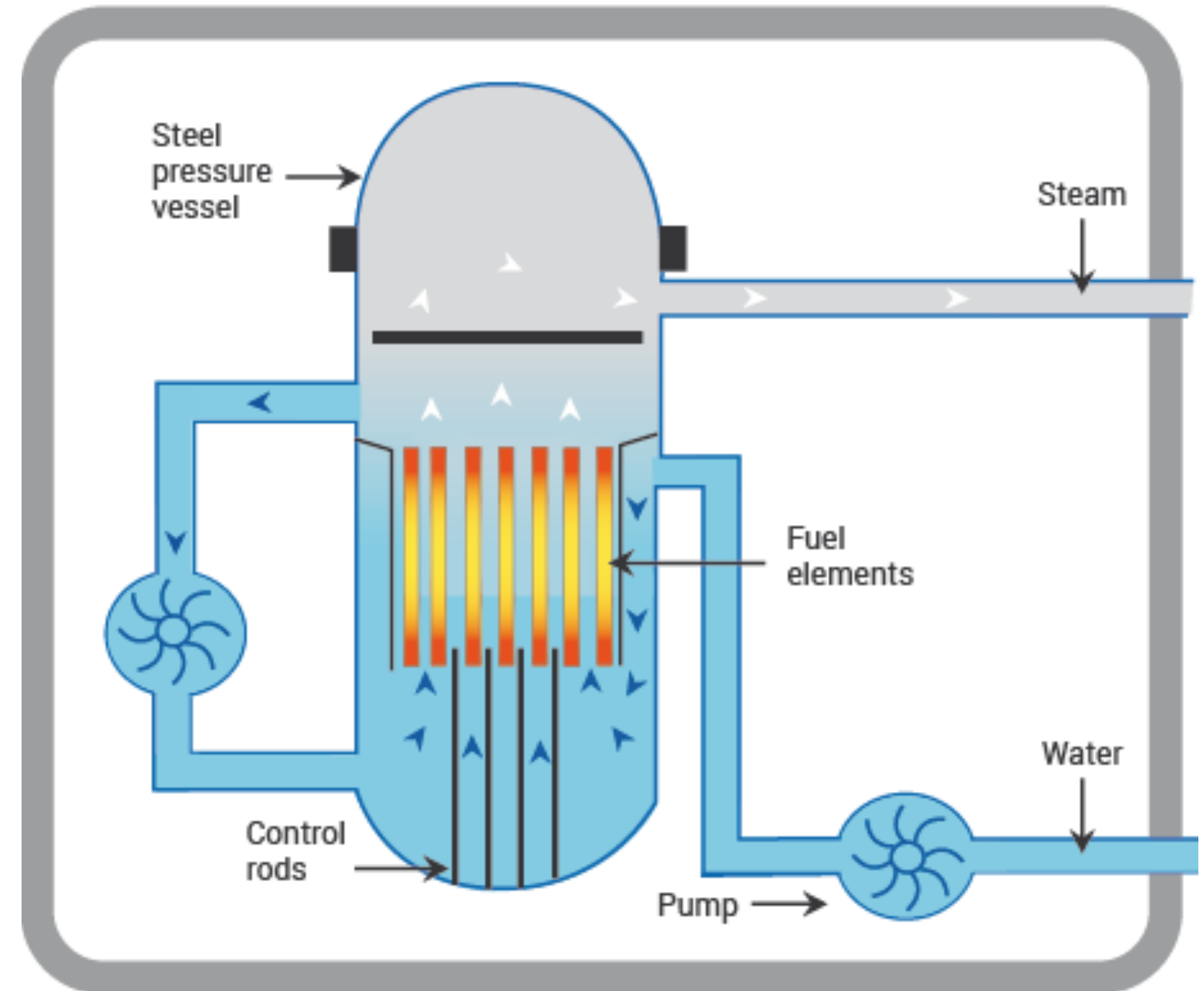
Reactor Designs

- Four reactor designs have been adopted for commercial use:
 - Boiling Water Reactors (BWR)
 - Pressurized Water Reactors (PWR)
 - Pressurized Heavy Water Reactor (PHWR)
 - Canada Deuterium Uranium (CANDU)
 - Reaktor Bolshoy Moschnosti Kanalniy (High Power Channel Reactor) (RBMK)

Uranium Fission Reactors

Boiling Water Reactor

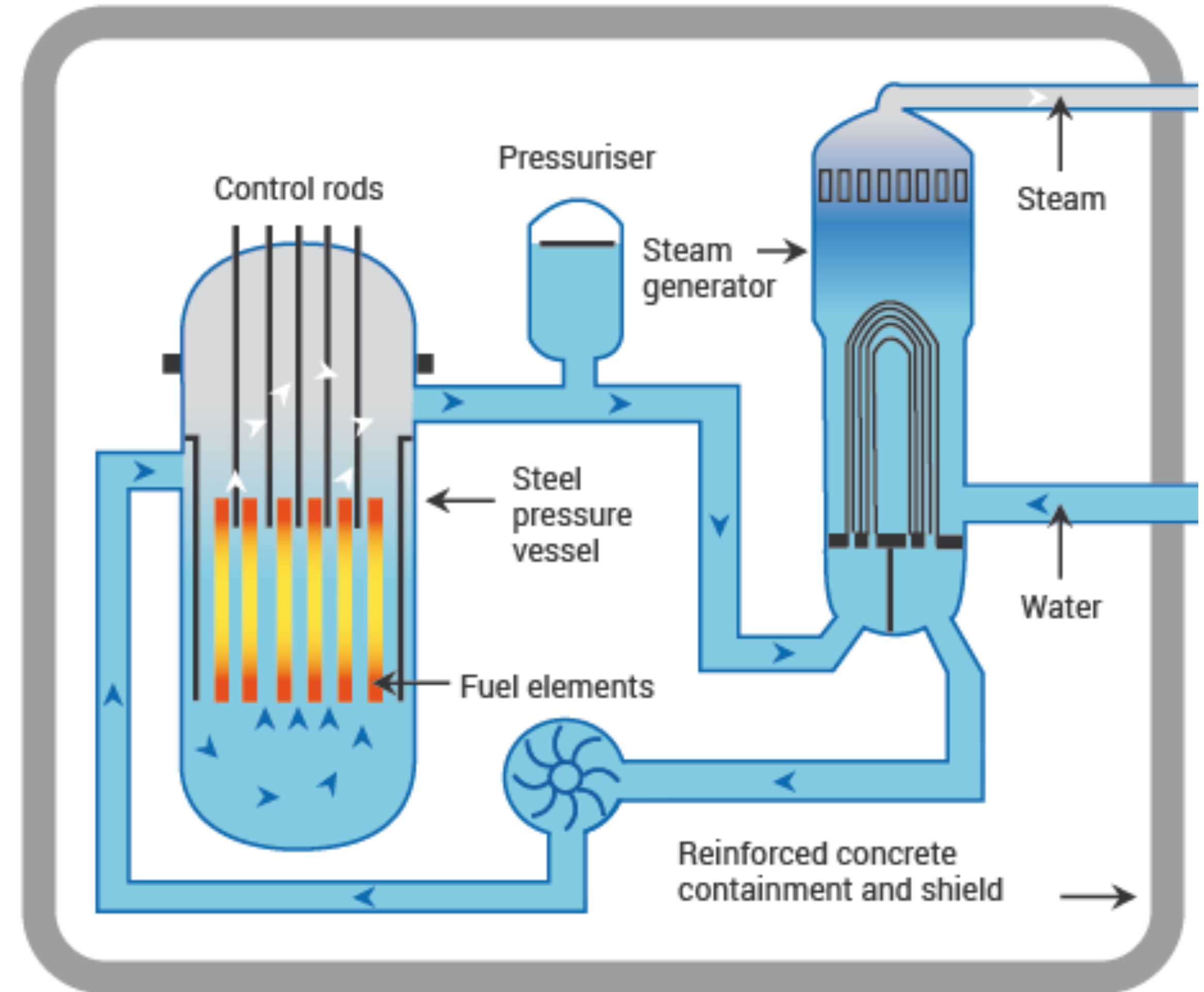
- Water is boiled directly in the core to produce steam
 - This water is both the coolant and the moderator
- Reaction rate is controlled by controlling the water velocity in the core and thus the steam bubble size
 - Bubbles reduce the density of the water and thus its moderation level



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Pressurized Water Reactor

- Reactor vessel is held under pressure to prevent water from boiling
- The use of a water to water steam generator means that no radioactive water enters the turbines
- PWRs can also be used to produce process steam



Uranium Fission Reactors

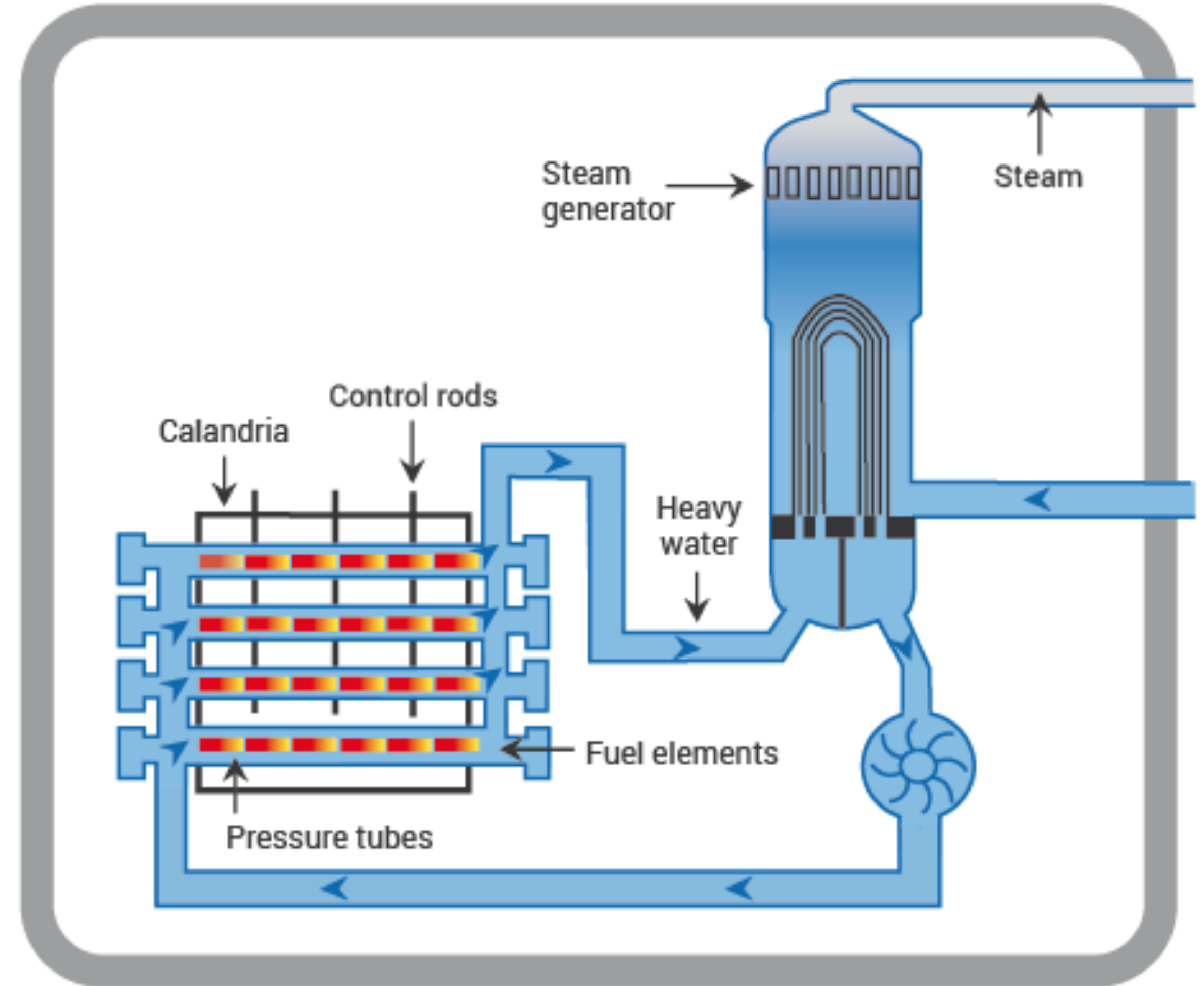
Pressurized Water Reactor

- Control rods are the only means of controlling the reaction
 - PWRs do not respond fast enough to be used for load following and are typically treated as base load sources
- Because only liquid water is present in the core, it can be significantly smaller than a BWR
 - This allows for a reinforced concrete containment structure / bio shield
 - PWRs are used for propulsion applications (boats / submarines)

Uranium Fission Reactors

PHWR / CANDU

- Heavy water (D_2O) is used as both a coolant and a moderator
- The calandria is a large vessel that contains heavy water for moderation
- Pressure tubes contain high pressure heavy water coolant as well as the fuel bundles



Uranium Fission Reactors

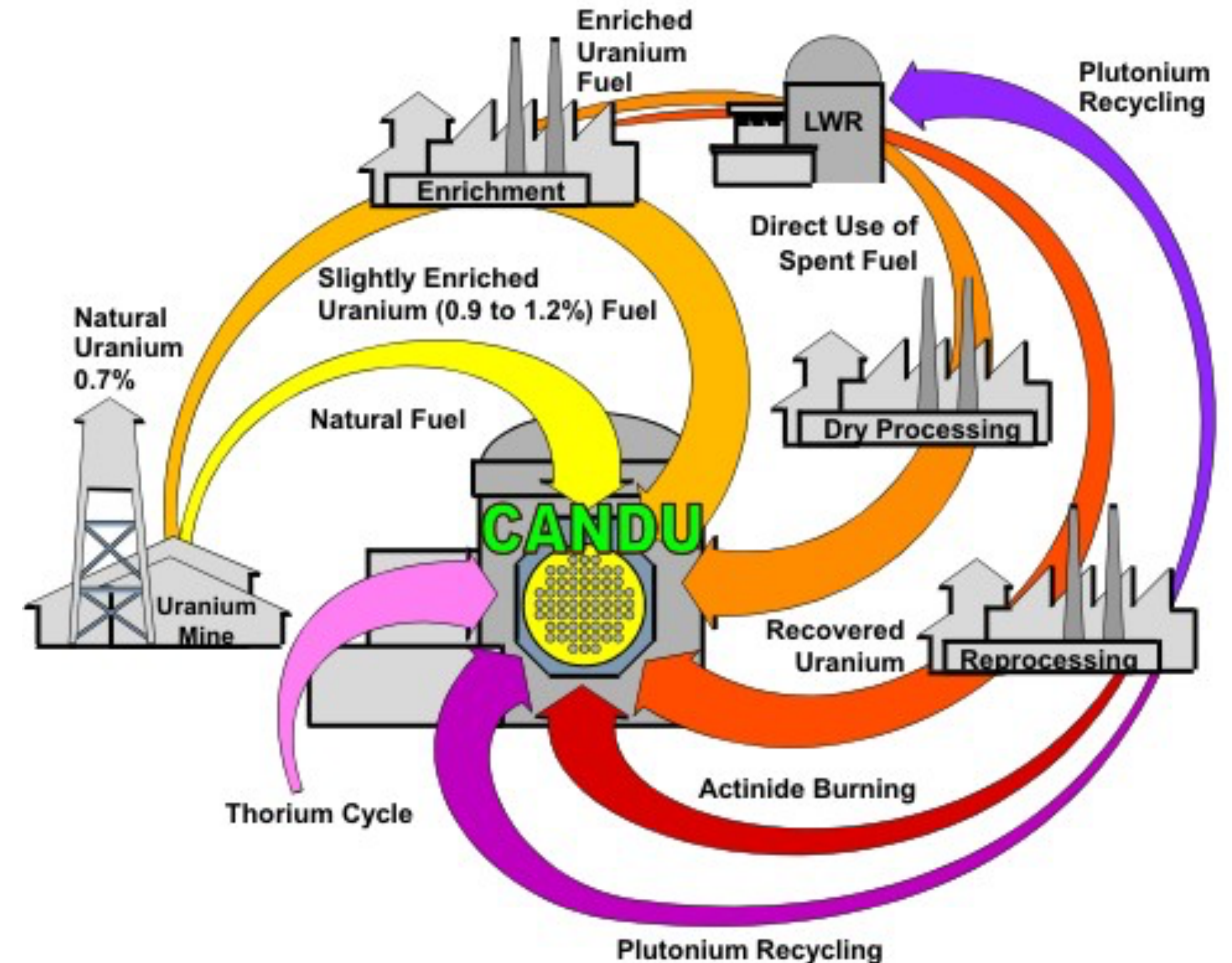
PHWR / CANDU - Heavy Water

- 99.75%+ Heavy Water (D_2O) is used as the moderator / coolant
 - In light water (H_2O) there is significant neutron loss to capture by the hydrogen in the water
 - The improved neutron efficiency allows for the use of unenriched fuel
- Natural water contains 1:3200 semi-heavy water (HDO) and 1:41 000 000 heavy water (D_2O)

Uranium Fission Reactors

PHWR / CANDU - Fuel Sources

- The reactor design was intended for natural uranium fuel (uranium oxide)
- Because of the high neutron efficiency of the design many fuel types can be used:
 - Natural or enriched uranium
 - Recovered or reprocessed uranium
 - Uranium / plutonium mix (MOX)
 - Thorium



Uranium Fission Reactors

Plutonium

- All uranium fission reactors produce some amount of plutonium as the result of neutron capture by ^{238}U
- In some reactors the fission of produced plutonium accounts for up to 1/2 of the energy generated
- Heavier than uranium elements account for some of long lived isotopes in reactor waste
- Breeder reactors are designed to maximize plutonium production

Uranium Fission Reactors

Void Coefficients

- How a reactor responds to a loss of coolant is called the void coefficient
- A typical minor loss of coolant is a steam bubble caused by boiling coolant
- The void coefficient depends on how the coolant is acting
 - If the coolant is acting primarily as a neutron moderator (light water in BWR or PWR) the void coefficient will be negative
 - a loss of coolant will stop or slow the reaction
 - This is used to control BWRs

Uranium Fission Reactors

Void Coefficients

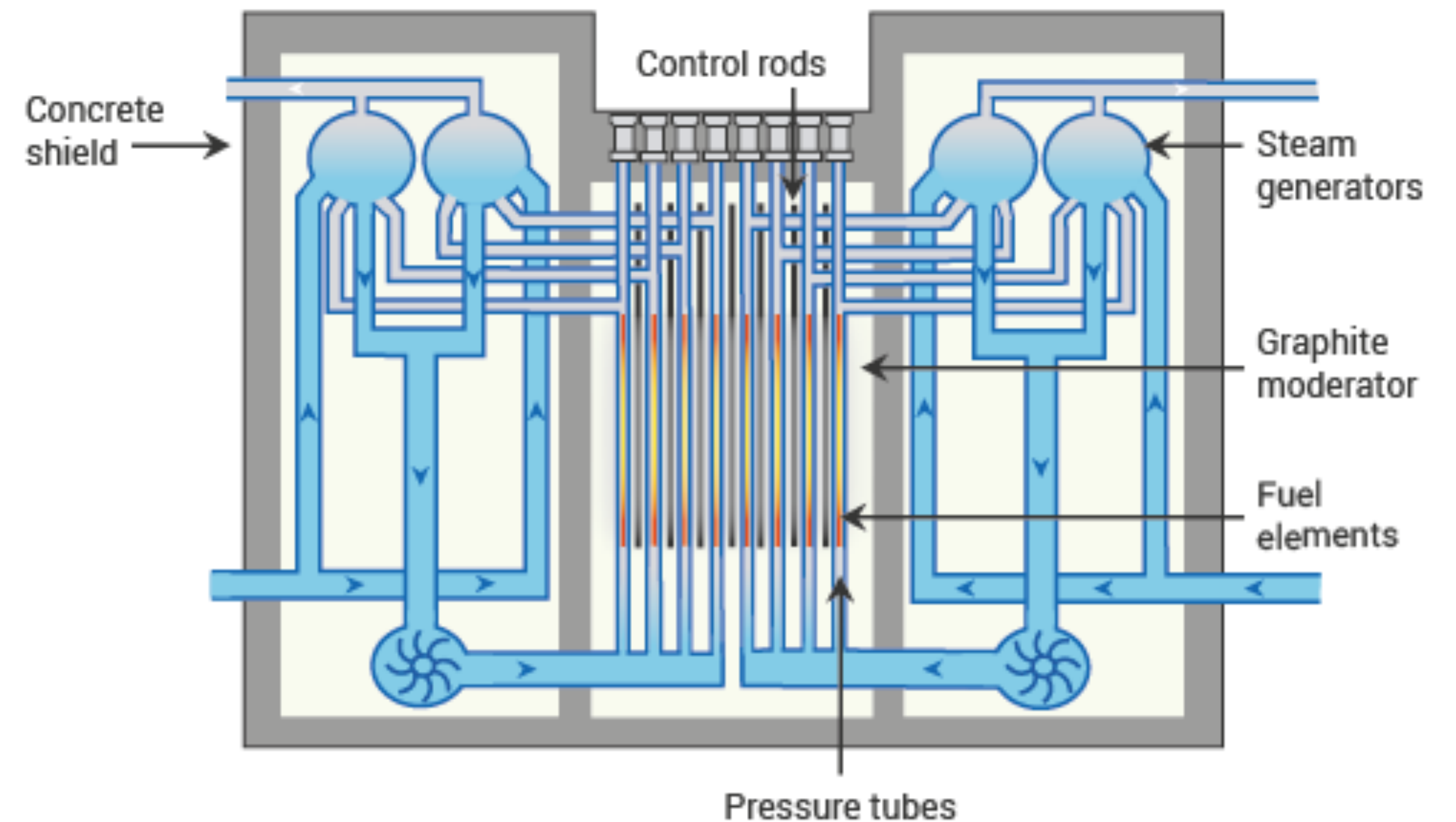
- If the coolant is acting primarily as a neutron absorber (this is typically the case when the moderator is separate from the coolant) the void coefficient will be positive
 - a loss of coolant will accelerate the reaction the reaction
 - This increases the risk of a meltdown
- The void coefficient of a PHWR is dependant on the fuel load
 - Natural uranium has a slight positive void coefficient

Uranium Fission Reactors

RBMK

- Designed to be used without the use of enriched fuel or heavy water
- Solid graphite acts as a moderator
- Natural water used as a coolant
- Modern installations use lightly enriched fuel

A Light Water Graphite-moderated Reactor (LWGR/RBMK)



Uranium Fission Reactors

Nuclear Meltdown

- A nuclear fuel assembly consists of small pellets of fissile fuel encased in metal tubes held in place by metal supports
- To maximize the presence of coolant / moderator a minimum of metal is used in the supporting structure
- A loss of coolant will lead to the melting of the fuel assembly



Uranium Fission Reactors

Nuclear Meltdown

- Melted fuel and other melted reactor components pool at the bottom of the reactor vessel
- If not cooled this material may melt through the reactor vessel and containment structure
- Hazards include:
 - Contamination of cooling water with fission products
 - Breach of containment structure - release of fission products into the environment
 - Steam explosion
 - Hydrogen explosion

Uranium Fission Reactors

Small Modular Reactors (SMR)

- An approach to reactor construction based on a smaller (typically <300mWe) reactor design
- Reactor is factory fabricated as modules and transported on site
- A wide range of reactor designs have been adapted to SMR
 - Most common are pressurized water
 - Ontario is constructing 4 BWRX-300 SMRs
 - A GE Vernova Hitachi boiling water design

Chernobyl

Chernobyl

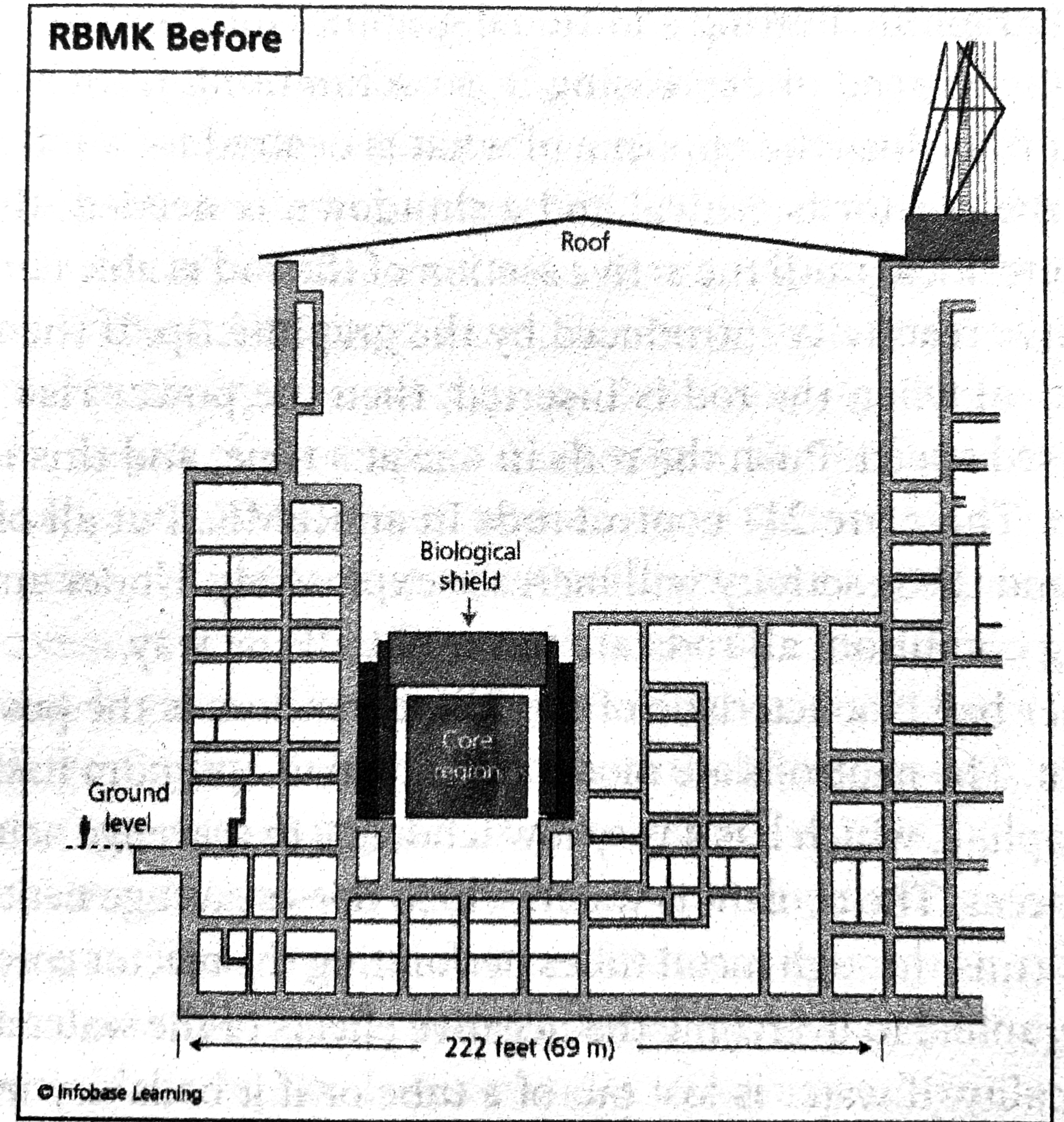
Background

- The RBMK reactor design was born out of a Soviet imperative to rapidly industrialize
- Design was intended to eliminate the need for costly uranium enrichment and heavy water
 - Neutron energy spectrum from graphite moderation is conducive to plutonium production
 - Graphite moderated boiling water design means the risk of both steam explosion and graphite fire

Chernobyl

Design Flaws

- Too large to build a single piece containment structure
 - Large reactor core 46' diameter 23' tall
 - Long fuel rods (12') require large equipment for online refuelling
- Slow emergency shutdown system (20s to deploy)



Chernobyl

Design Flaws

- Control rods are graphite tipped (as a lubricant) followed by a 3.23' of hollow tube
 - Inserting a control rod initially adds extra moderator with some delay before the neutron absorber enters the core
 - Hollow control rod sections displace water (a weak neutron absorber) from the core
- Water bath under core to dilute any leaks from failing cooling pipes
 - If the core melts down and falls into this bath it will flash into steam
 - The resulting steam explosion would be directed upwards towards the reactor core
- Positive Void Coefficient

Chernobyl

What Went Wrong

- To test a backup system that used the inertia in the turbines to power cooling pumps a control system was installed to create a worst case shutdown
 - Turbine shutdown
 - Disable ECCS (emergency core cooling system)
 - Disable all pumps
 - Disable diesel backup generators
- Automatic reactor control disabled for test
 - Operators were having trouble manually balancing the reaction across the core

Chernobyl

What Went Wrong

- Prior to the test the reactor entered an 'iodine valley'
- To restore power operators had fully removed 205 or 211 control rods
 - because of the control rod design this condition risked the rapid insertion of all rods (a rapid shutdown) causing a super-criticality and steam explosion
- Ignoring a system warning that reserve reactivity was too low and the reactor should be shut down, operators proceeded with the test

Chernobyl

What Went Wrong

- With the loss of power demand from the turbine, the cooling water in the reactor boiled resulting in an increase in reactor power
- Operators activated the emergency shutdown system
- The control rods jammed due to twisted guide pipes after moving only about 6.5' into the core
- Heat from the rapidly accelerating reaction caused a steam explosion destroying much of the cooling system

Chernobyl

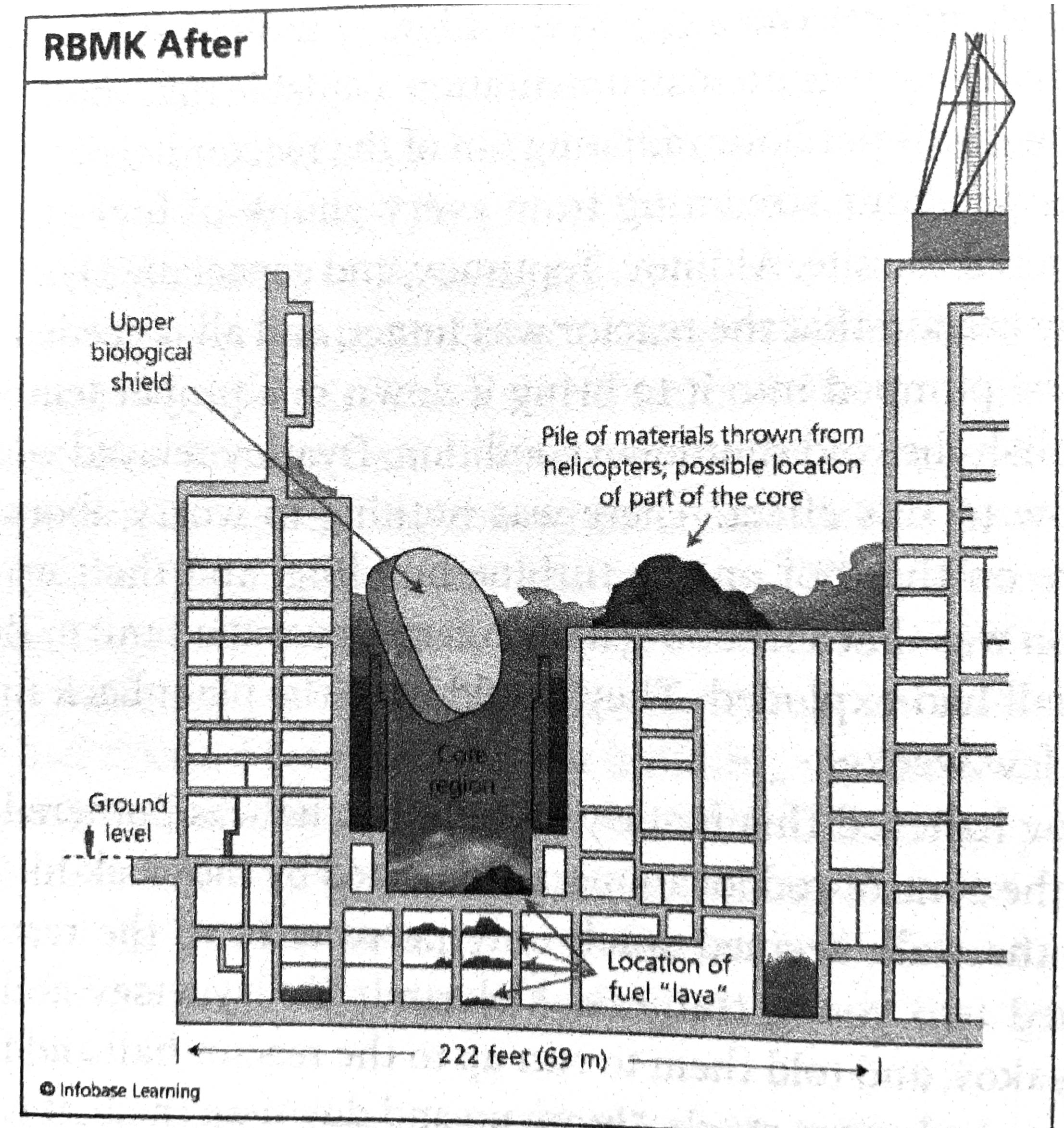
What Went Wrong

- Melted zirconium reacted with water to release hydrogen gas
- The steam explosion removed the bio-shield from the top of the reactor
- Exposed to the atmosphere the hydrogen exploded
- Aerosolized fuel (~1/2 of fuel load) and reactor core components were propelled into the atmosphere
- The explosion started numerous fires at the plant including a graphite fire in the core

Chernobyl

What Went Wrong

- Radioactive fallout covered all of Europe (including Scandinavia and the UK)
- Firefighting efforts (using air dropped sand, clay, lead, and boric acid) insulated the remaining, still hot, fuel
- Liquid fuel dripped into basement of reactor building



Fukushima

Fukushima

Background

- In 2010/11 approximately 40% of Japan's electricity is generated by 54 reactors
- Japanese reactors all located on the coast and use sea water cooling in place of cooling towers
 - Plants in north-eastern Japan are on the Pacific coast near the Pacific plate subduction zone
 - This coast is prone to tsunamis
 - Fukushima 1 brought online March 26, 1971
 - General Electric BWR/3 with Mark I containment
 - Final reactor (Fukushima 6 GE BWR/5) brought online October 1979

Fukushima

Background GE BWR Design

- To keep the reactor design simple and minimize piping the steam separators and driers are located on top of the reactor vessel
 - The reactor is 60' tall
 - A one piece concrete containment building cannot be built this large
 - Instead these reactors used a steel containment 'dry well' with a bolt on lid
 - This containment cannot withstand a steam explosion so it is piped into a 'wet well' to soak and cool any steam present

Fukushima

Background GE BWR Design Flaws

- In the 1970s a German plant operator had concerns about the Mark 1 containment design tested it and found that the wet well could not withstand a full force steam discharge
- In 1980 GE issued a set of modifications to the containment
 - These were completed by all operators
- Criticism of the Mark 1 containment continued

Fukushima

Background GE BWR Design

- The Fukushima 1 reactor was equipped with an isolation condenser
 - Steam flow to the turbine would be shutdown during an emergency shutdown
- Reactors were also equipped with a High Pressure Coolant Injection system (HPCI)
 - The HPCI pump is steam powered using the reactor as a source
- Reactor steam relief valve opens into the wet well in the case of excess pressure
- A pressure relief in the wet well dissipates steam to the environment

Fukushima

Background GE BWR Design - Electrical

- All valves in the plant require electricity to operate
- In the case of an emergency shutdown the turbine generator is shut down
- Plants were cross wired to a neighbouring plant for backup power
- 2 redundant diesel generators provide additional backup
- Lead acid battery banks provide an additional layer of backup

Fukushima

The 2011 Tohoku Earthquake

- The plants were initially hit by a magnitude 9.0 earthquake
 - The earthquake produced 0.561g, the plants were designed for 0.447g
 - External power lines went down
 - Some pipe-runs were damaged
 - Plants 1 through 3 shut down automatically
 - Emergency power was working correctly
 - Plants 4 through 6 were already shut down and in various stages of refuelling
- An operator at unit 1 decided that the isolation condenser was cooling the reactor too quickly and shut it down

Fukushima

The 2011 Tohoku Tsunami

- 3 tsunami waves hit the beach after the earthquake, the second and third waves overwhelmed the tsunami wall and flooded the plant
 - The waves damaged the sea water intakes and pumps for all plants
 - The generators and battery banks for units 1 and 2 (in the basement) flooded and these plants were without any instrumentation or control
 - Units 3 and 4 were operating on battery power
 - Operators set about making sure system could run passively when the battery run out
 - Units 5 and 6 were running on diesel generators

Fukushima

The 2011 Tohoku Tsunami

- Unit 1 was now without power with the residual heat in the core and all cooling systems shut off
- Units 2 and 3 would eventually need AC power because the water in the wet well would become too hot to effectively cool the reactor
- Portable generators were available but could not be brought on site due to road damage

Fukushima

Unit 1 Meltdown

- Coolant boiled out of the core in unit 1 and fuel started to melt
 - The zirconium fuel cladding reacted with the steam to oxidize and form hydrogen
- The steam and hydrogen pressure exceeded the capacity of the containment
 - With no electricity to operate relief valves, the containment ruptured releasing radioactive steam and hydrogen into the reactor building
- The unit 1 reactor building exploded releasing radioactive dust
 - The resulting site contamination made work on site more complicated

Fukushima

Unit 3 Meltdown

- The day after the tsunami the HPCI on unit 3 stopped working
- Following a similar path to unit 1 there was a core melt down and hydrogen gas explosion

Fukushima

Unit 2 Meltdown

- Later the day after the tsunami the HPCI on unit 2 stopped working
- Following a similar path to unit 1 there was a core melt down
 - Due to the hole in the reactor building from the unit 1 explosion, the hydrogen dissipated preventing a 3rd explosion
 - Radioactive iodine and xenon were dissipated to the environment

Fukushima

Unit 4 Explosion

- With no power to operate them the steam relief valves on unit 4 were left open
 - When unit 3 was vented to the shared stack, 1/2 of the radioactive steam and hydrogen entered the unit 4 wet well
- 4 days after the earthquake, there was a hydrogen explosion at unit 4

Fukushima

Consequences

- None of the waste stored on site was released
- The 3 running reactors all experienced meltdowns
 - Radioactive material was released from the fuel loads due to the containment structure's design not anticipating prolonged power outage
- An area 10 x 40 km is heavily contaminated with fallout and uninhabitable
- Rainstorms wash fallout into the pacific

Fukushima

Consequences

- The Fukushima II plant located 11 km down the coast was hit by effectively the same earthquake and tsunami
 - 4 GE BWR/5 reactors with Mark II containment structure online 1981-1986
- While the plant sustained damage it was able to be safely shutdown and later restarted
 - The electrical switch yard was undamaged allowing interconnecting of plants to share backup generators
- TEPCO decided to decommission the plant in 2019

Three-Mile Island

Three-Mile Island

Background

- Two PWRs on a sand bar in the Susquehanna river
- Built by Babcock and Wilcox (B&W)
- Owned by Metropolitan Edison of Pennsylvania

Three-Mile Island

What Went Wrong

- The Pilot Operated Relief Valve (PORV) had been leaking steam from the primary cooling loop into the containment building
- An airline used to flush resin beads used to clean the water in the steam loop had been left open
 - Due to insufficient air supply in the general air system, the instrumentation air system had been interconnected to the general air system at the resin tanks
 - Due to a faulty check valve water entered the instrumentation air system

Three-Mile Island

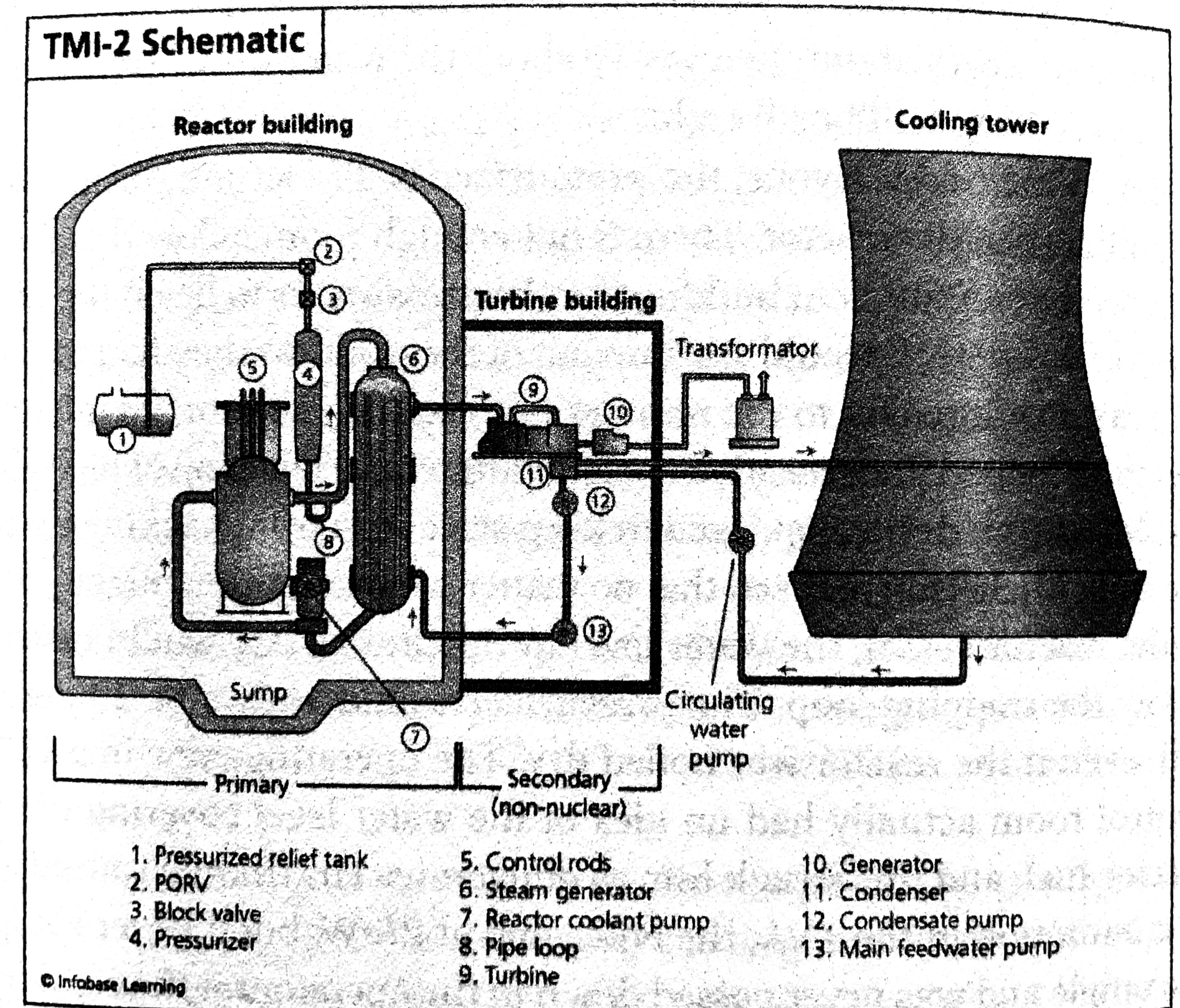
What Went Wrong

- The water backup in the steam system caused all valves on the secondary cooling loop to close at the same time
 - The resulting water hammer severely damaged piping
- Within seconds of the loss of the secondary cooling loop, the turbine and reactors both automatically shut down
- Due to the loss of cooling the pressure in the primary cooling loop rose
 - The PORV opened for 10s to relieve excess pressure in the primary loop
- 2 minutes later the Emergency Core Cooling System (ECCS) activated

Three-Mile Island

What Went Wrong

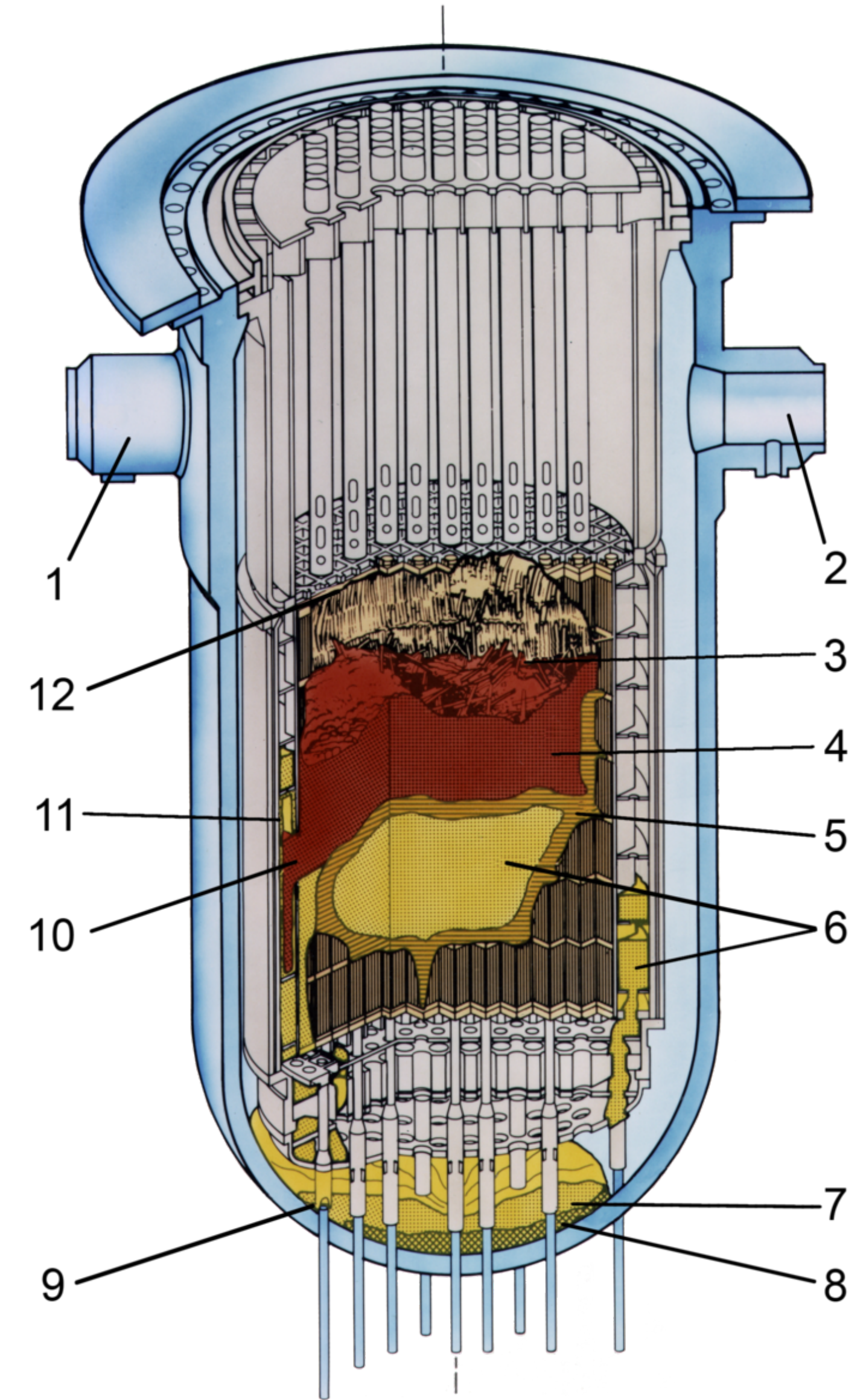
- The pressurizer water level continued to rise and the primary loop pressure continued to drop
- Water had boiled out of both steam generators
- The ECCS was shutdown to prevent the pressurizer from filling with liquid water



Three-Mile Island

What Went Wrong

- Unbeknownst to the operators, the PORV was stuck open, accounting for the falling pressure
 - Because of the low pressure, the water in the core was able to boil
 - Due to a lack of instrumentation the control room had no indication of the water level in the main reactor vessel
- As the water boiled below the level of the fuel rods, the fuel rods began to melt



Three-Mile Island

What Went Wrong

- After the relief tank filled with water, it began to spill on to the containment building floor and collect in the sump
 - Pumps in the sump transferred the radioactive coolant to an un-sealed auxiliary building where it was able to be contained
- More than 2 hours after the loss of coolant the order was given to shut the block valve on the PORV
 - Had this been done early it would have prevented the boiling of coolant in the core and resulting meltdown

Three-Mile Island

The Outcomes

- Nearly everything that could go wrong did
- Offsite radiation levels peaked at 0.007 rem/h (civilian max does is 0.5 rem/a)
- Discharge was primarily non bioactive ^{133}Xe
- No one harmed during the incident
- No increase in cancer rates attributable to the incident
- Containment structure remained intact
- TMI2 reactor is not operable or repairable

Three-Mile Island

Lessons Learned

- Many changes in instrumentation and control design as well as operator training
- 35-40% of fuel melted
- 70% of core structure collapsed
- Meltdown had formed a layer of zirconium and uranium oxides on the reactor floor that would prevent the core from melting through the bottom of the reactor vessel
- TMI1 continued to operate (shut down in 2019)
 - Efforts to restart TMI1 are currently underway for a start date as early as 2027